

Personal AI Clone — Sprint 3 Report

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Section: C

Timeline: Nov 15 – Nov 28, 2025

Overview

Sprint 3 focused on improving the personalized AI model's performance, adding advanced features, integrating with a UI, performing safety & evaluation enhancements, and preparing deliverables for submission.

Key Tasks & Workflow

1. Dataset Expansion & Augmentation

- Added new user conversation data (approved only).
- Performed text normalization and augmentation (paraphrasing, synonyms).
- Re-generated JSONL datasets for training and validation.

Files:

- `data/processed/user_corpus_aug.jsonl`
 - `data/processed/user_corpus_val_aug.jsonl`
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2. Model Enhancements

- Fine-tuned the LoRA adapter with augmented dataset.
- Tested different hyperparameters for improved perplexity and output diversity.
- Saved updated model to `models/user_clone_v2/`.

Files:

- `training/train_v2.py`
 - `logs/train_run_v2_YYYYMMDD.txt`
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3. Advanced Evaluation

- Compared baseline GPT-2 vs personalized LoRA model.
- Computed embedding similarity using SBERT for 30 prompts.
- Conducted human evaluation with 3 reviewers on 15 prompt pairs.
- Recorded results in `docs/eval_examples_v2.md`.

Files:

- `docs/eval_examples_v2.md`
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4. Frontend Integration

- Updated `ui.html` to interact with new model version.
- Added user-friendly controls:
 - Max tokens
 - Temperature slider
- Configured CORS and proxy settings for local testing.

Local UI URL:

- <http://127.0.0.1:3000/ui.html>

Files:

- `frontend/ui.html`
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5. Safety & Guardrails

- Enhanced toxicity checks with updated Detoxify thresholds.
- Logged all generated outputs securely in encrypted audit logs.
- Ensured human-in-the-loop mechanism for sensitive outputs.

Files:

- `backend/safety.py` (updated)
 - `backend/model_server.py` (integrated safety features)
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6. Documentation & Reports

- Updated training report with:
 - New dataset stats (#examples, #tokens)
 - Updated hyperparameters
 - Training/validation loss graphs
 - Sample outputs and evaluation metrics

Files:

- `docs/training_report_v2.md`
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7. Final Deliverables

- All model files saved to `models/user_clone_v2/`.
- Committed frontend, backend, and documentation.
- Tagged release for Sprint 3:

```
git tag -a sprint3 -m "Sprint 3 complete: model improvements & frontend integration"
git push origin sprint3 --tags
```

